

GRMHD Tutorial Day 1

Ben Prather,
Black Hole Initiative Fellow, Harvard University

Instituto de Astrofísica de Andalucía,
Granada, Spain

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Notes for This Week

Lesson Plan

Morning Lectures:

- 1.
- 2.
- 3.
- 4.
- 5.

Lesson Plan

Morning Lectures:

1. What is GRMHD and why do we use it?
- 2.
- 3.
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3. What else do you need in a GRMHD algorithm?
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3. What else do you need in a GRMHD algorithm?
4. KHARMA in particular
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3. What else do you need in a GRMHD algorithm?
4. KHARMA in particular
5. Beyond GRMHD

Lesson Plan

Tutorials:

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Lesson Plan

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1. Getting KHARMA running; first simulation
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4. Hike at Dehesa del Generalife!
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Tutorials:

1. Getting KHARMA running; first simulation
2. Shock tubes, hydrodynamics, instabilities
3. Magnetohydrodynamics, first tori
4. Hike at Dehesa del Generalife!
5. Measuring simulation properties

NB

1. This is an *outline*!
2. Please feel free to change things up: ask what *you'd* like to know!
3. This timeline is *optimistic*. We don't have to cover everything this week.

Part 1: What use is GRMHD?

What isn't GRMHD?

GRMHD isn't flawless!

-
-
-

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What isn't GRMHD?

GRMHD isn't flawless!

- System is highly unstable
- Can be inaccurate, wrong about details
- Some elements always determined by floors

What isn't GRMHD?

"All simulations are wrong. Some are useful."

— Loose paraphrase of a saying among statisticians, attributed to George Box

What is GRMHD?

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4. Which does not emit or absorb radiation (**Non-radiative**)

What is GRMHD?

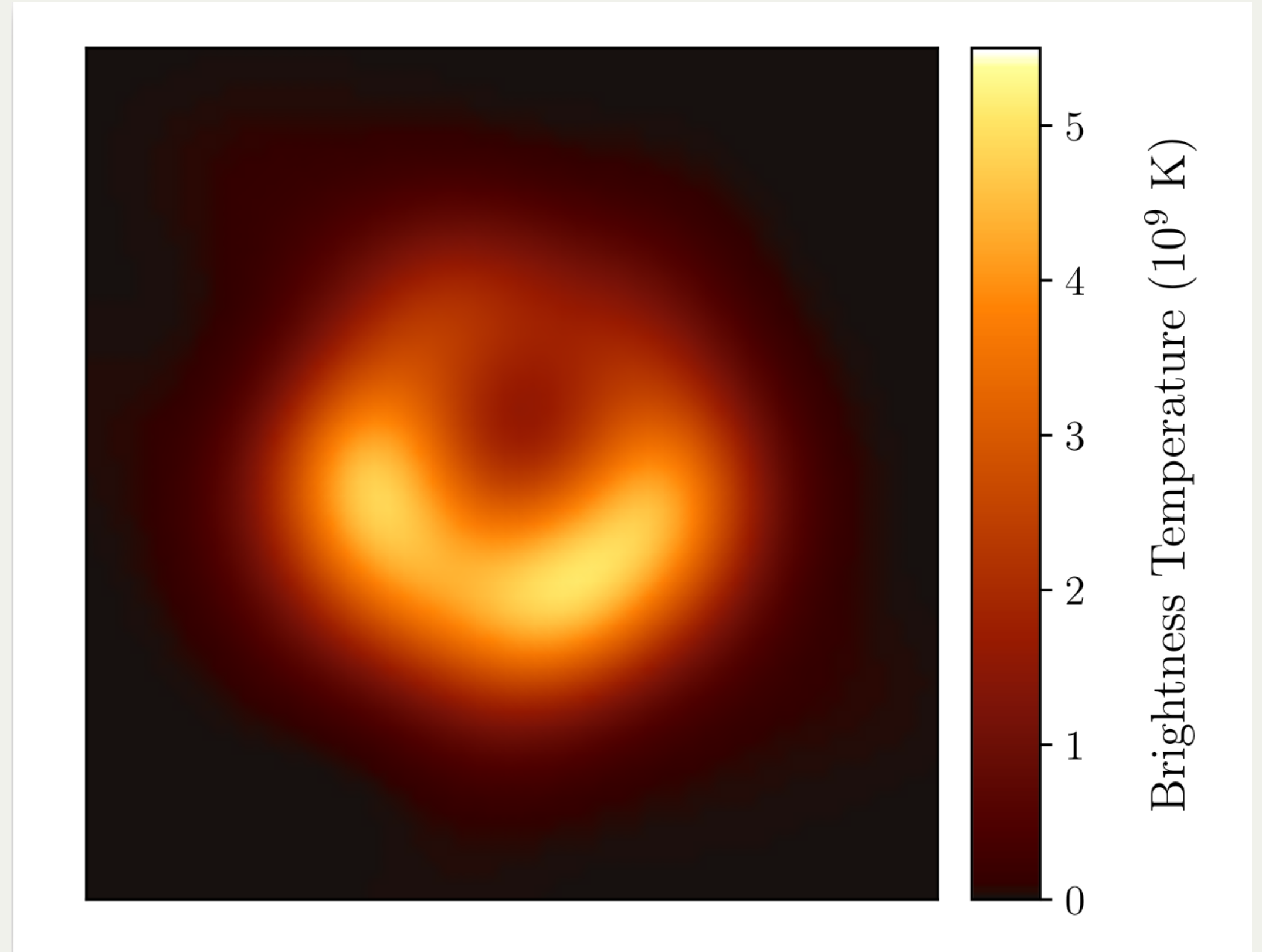
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 - a. And thus a "frozen-in" magnetic field (**Flux-freezing**)
4. Which does not emit or absorb radiation (**Non-radiative**)
5. Or exert meaningful self-gravity (**Non-gravitating**)

What is GRMHD used for?

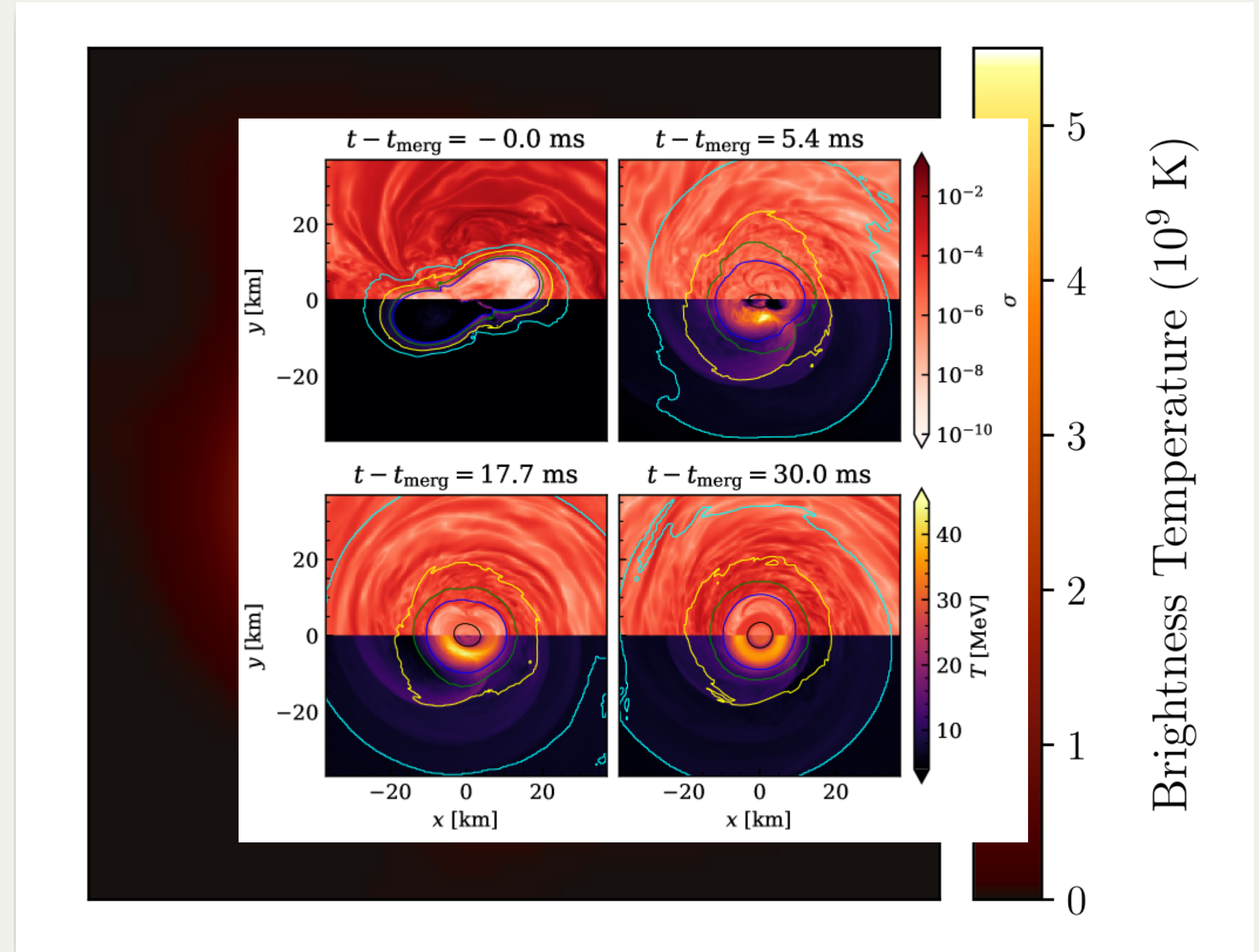
What is GRMHD used for?

- Black hole accretion



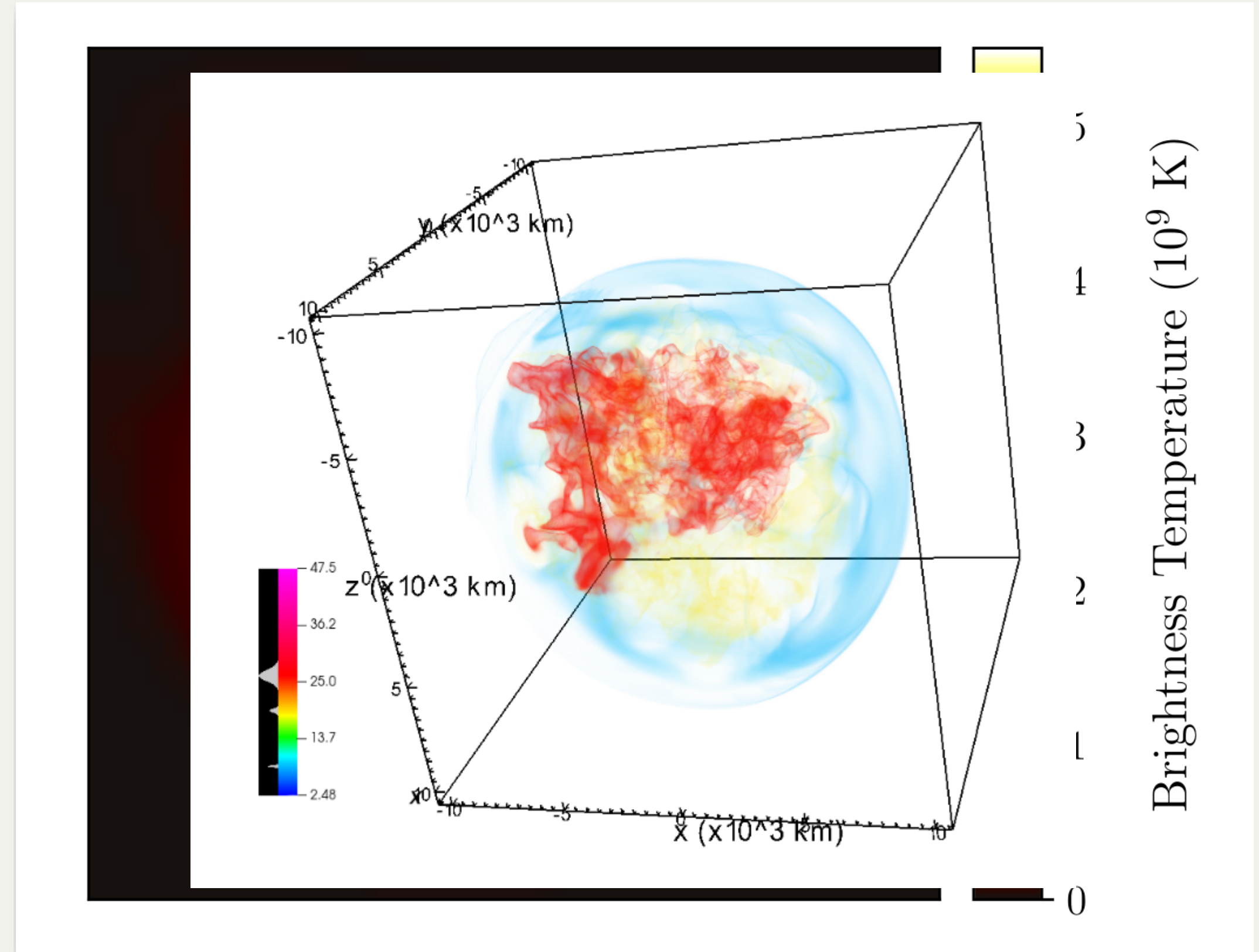
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- Neutron Stars, Kilonovae



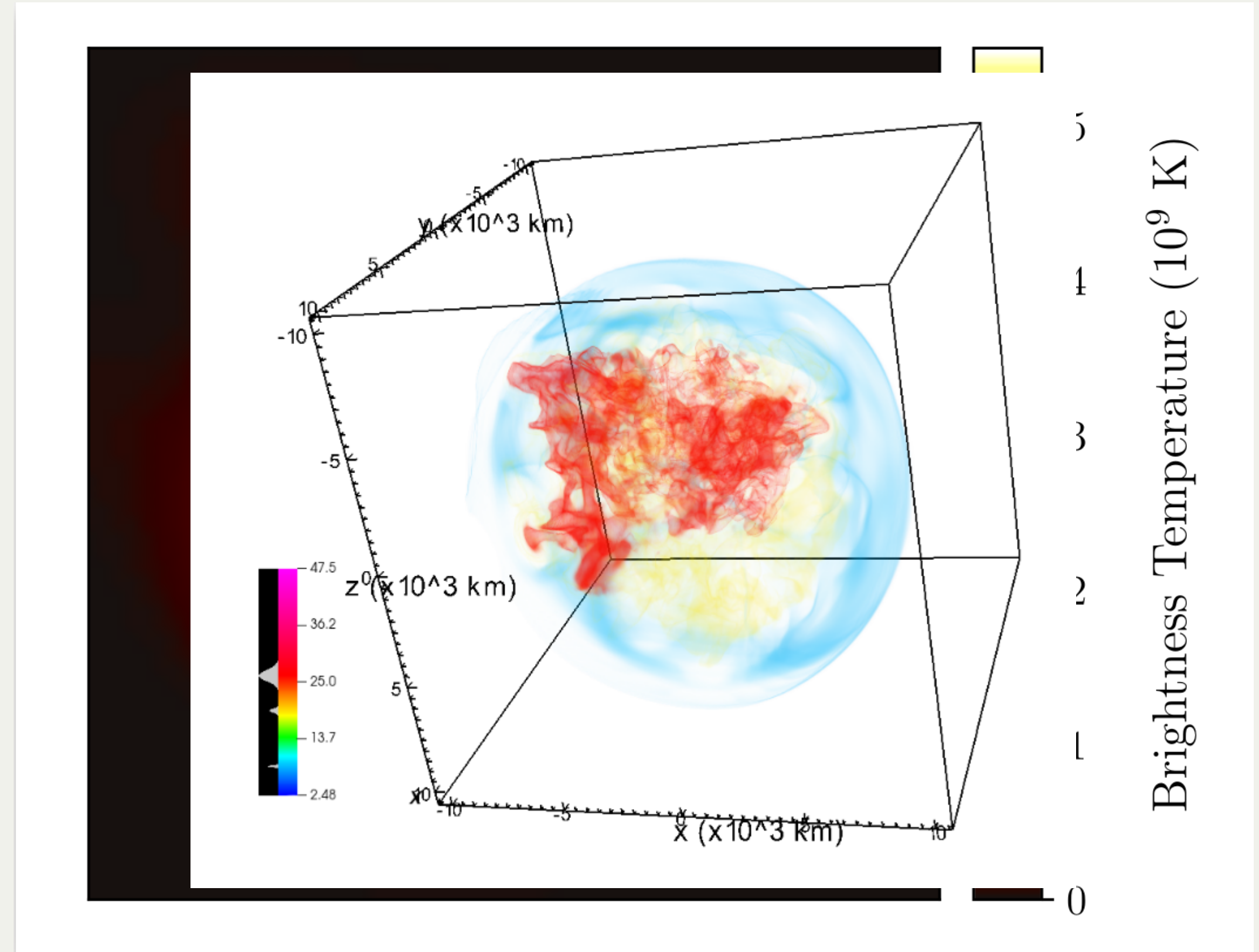
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- Black hole accretion
- Neutron Stars, Kilonovae
- Relativistic Supernovae



What is GRMHD used for?

- Black hole accretion
- Neutron Stars, Kilonovae
- Relativistic Supernovae
- Anything **large & dense** (LRDs?)



What isn't GRMHD used for?

The GRMHD we'll be doing is not for:

- Non-relativistic systems
- Dense/incompressible systems
- Sparse/kinetic systems
- Self-gravitating systems
- Highly resistive or viscous systems
- Radiatively efficient or mostly radiation-pressure systems

What GRMHD codes are there?

GPU/Hybrid:

- KHARMA
- H-AMR
- AthenaK

CPU:

- iharm3d
- BHAC
- KORAL
- Athena++
- Many other HARM-like...

Dynamical Spacetimes:

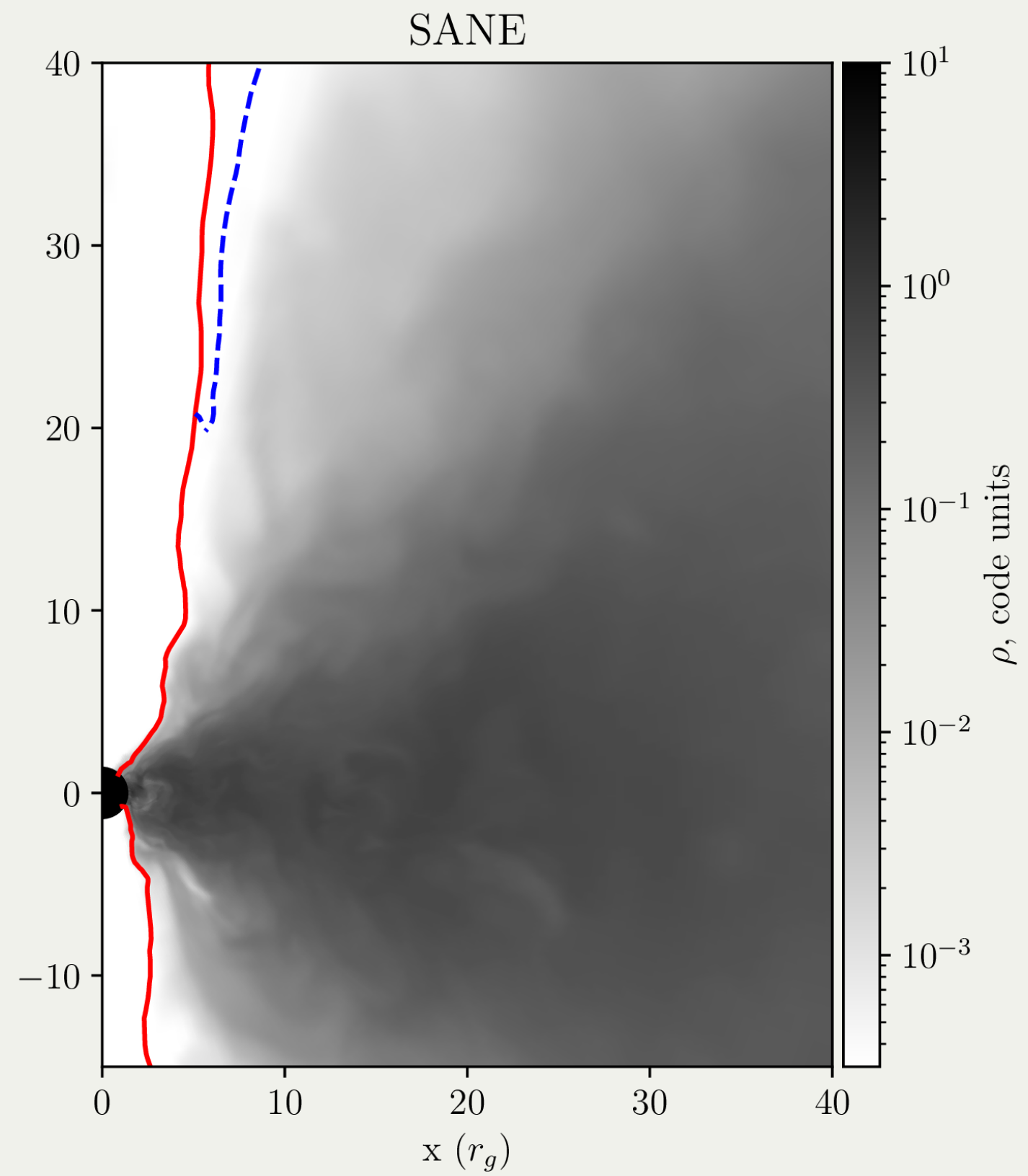
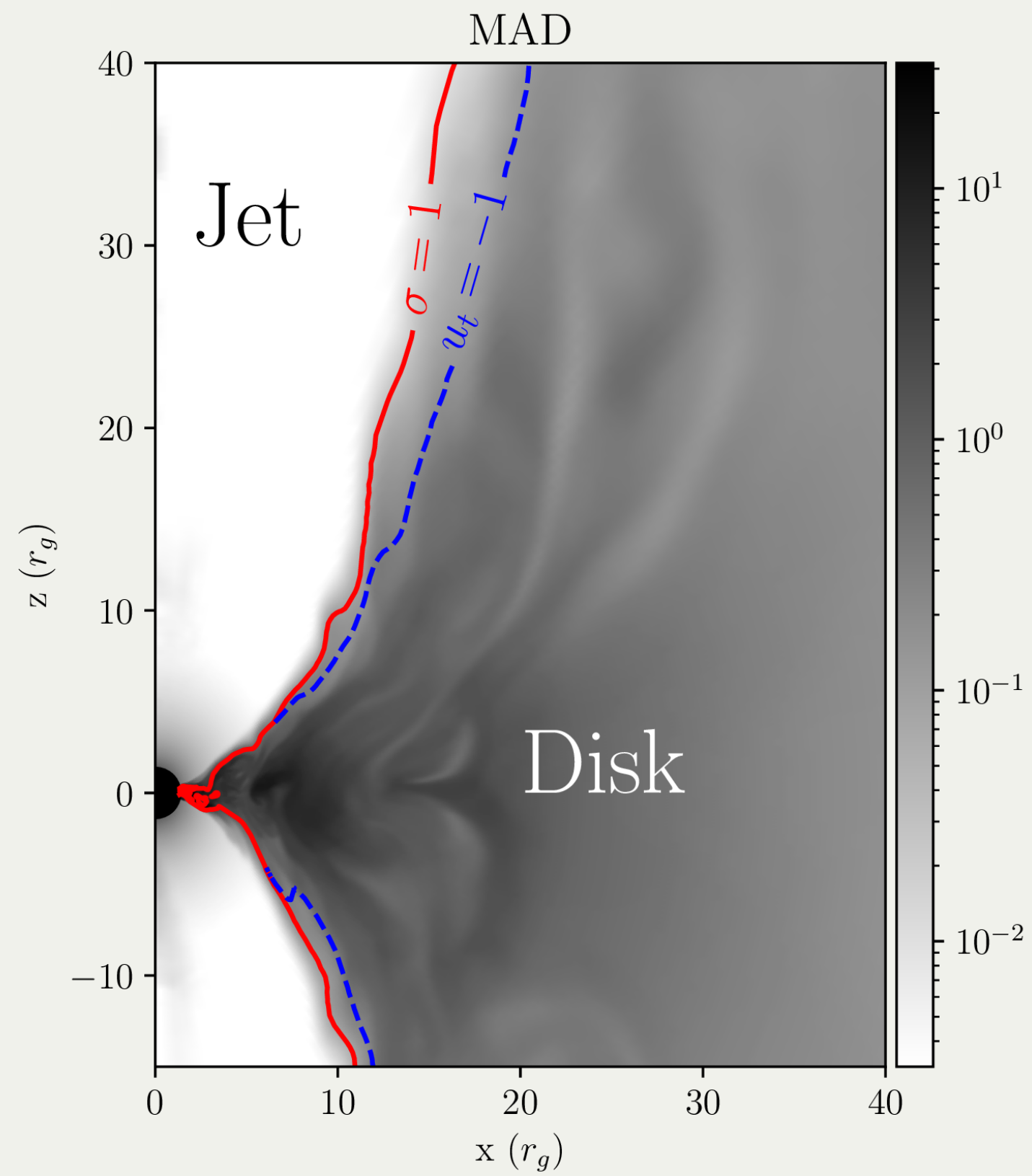
- Phoebus
- IllinoisGRMHD
- Whisky/WhiskyTHC
- Spritz
- Many others

Getting matter to the EH

Must transport energy, angular momentum away

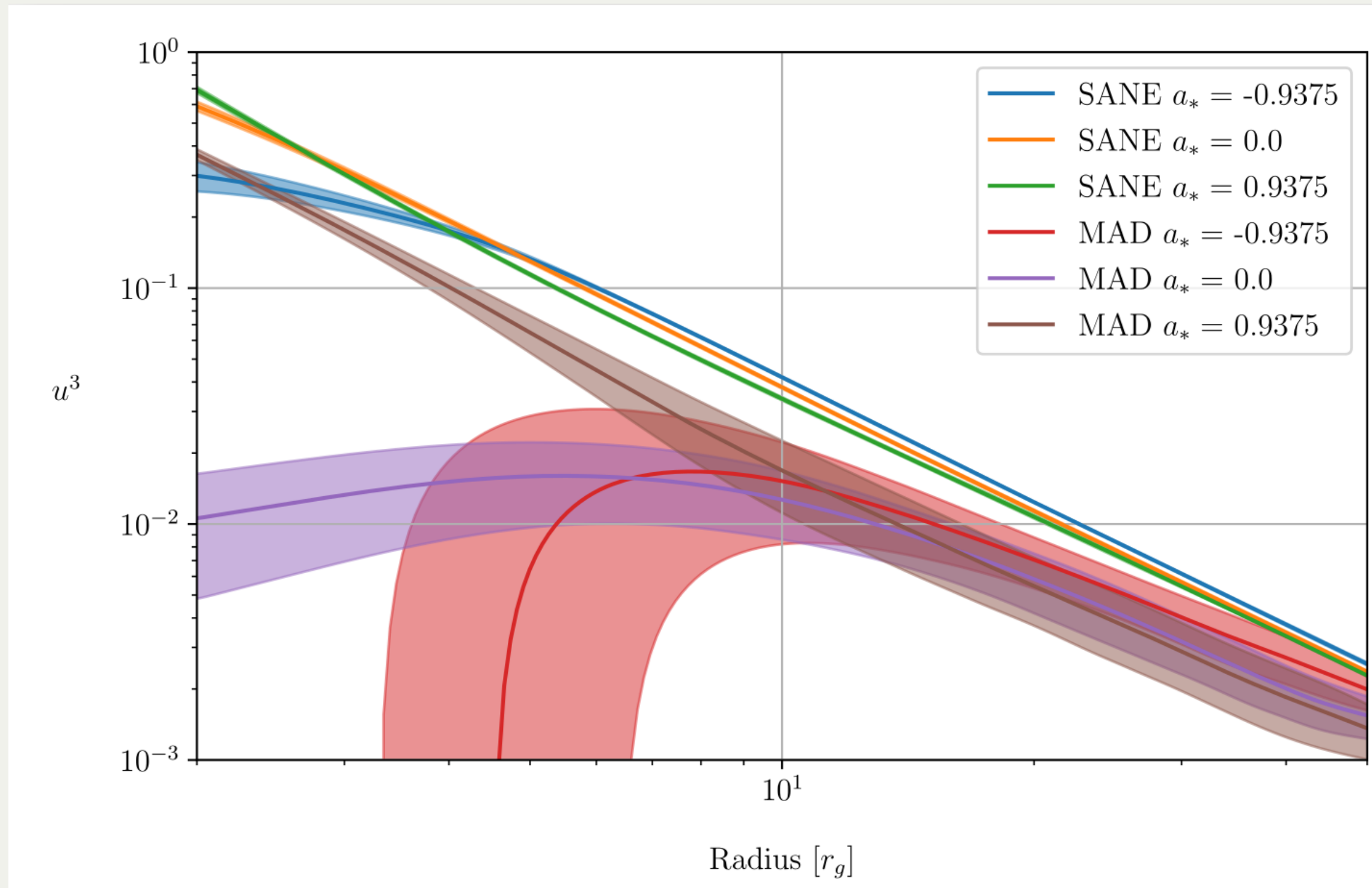
- Radiation (thin disk)
- Viscosity (alpha-disk, ADAF, RIAF)
- Magnetic field can provide "viscosity" efficiently

Disk Profile



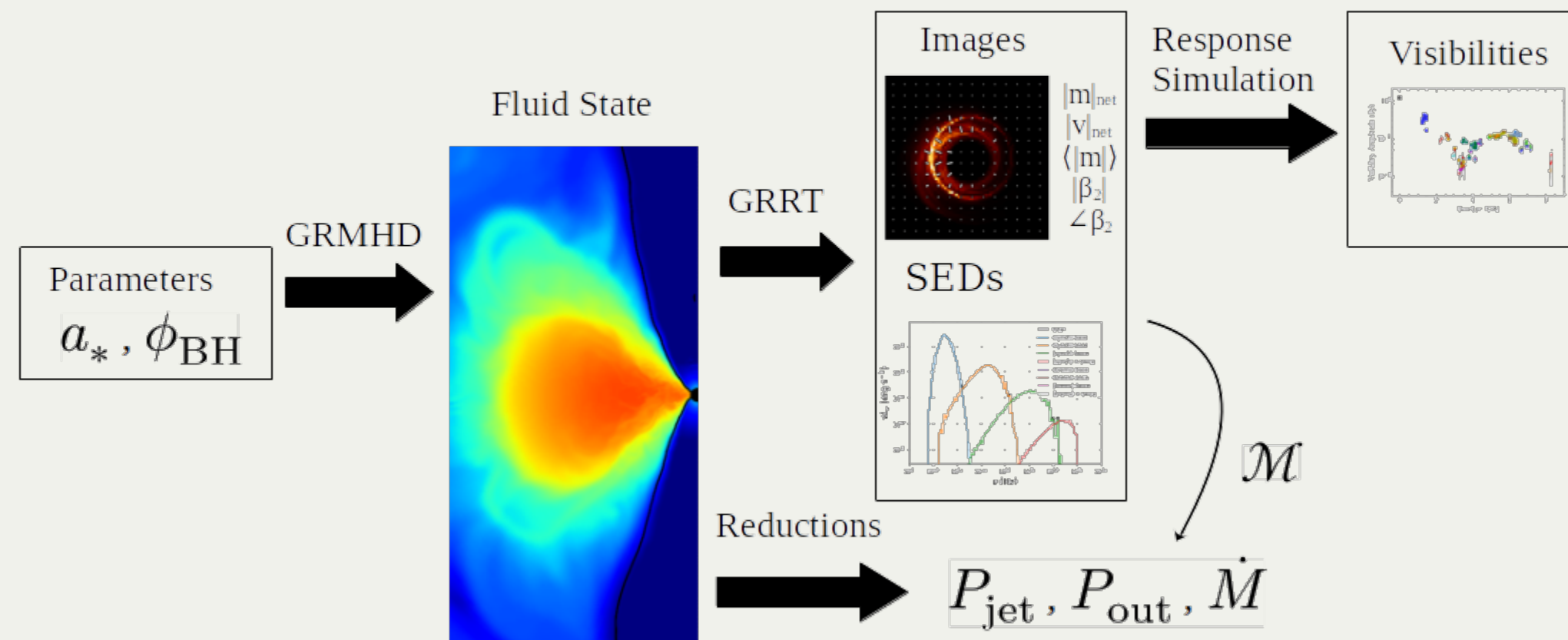
MADs and SANEs

Rotation Profiles



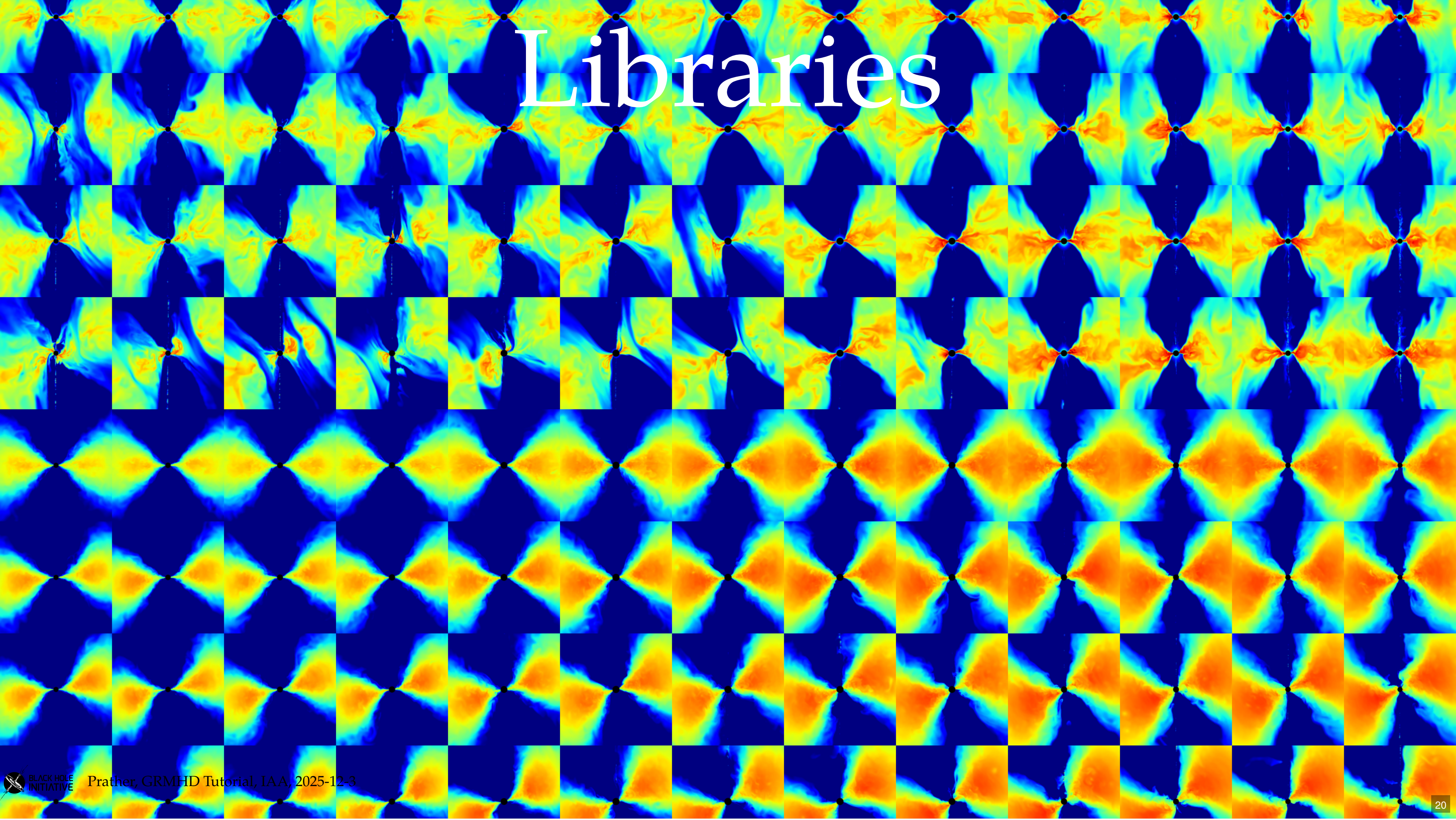
GRMHD in EHT

The EHT Theory Pipeline

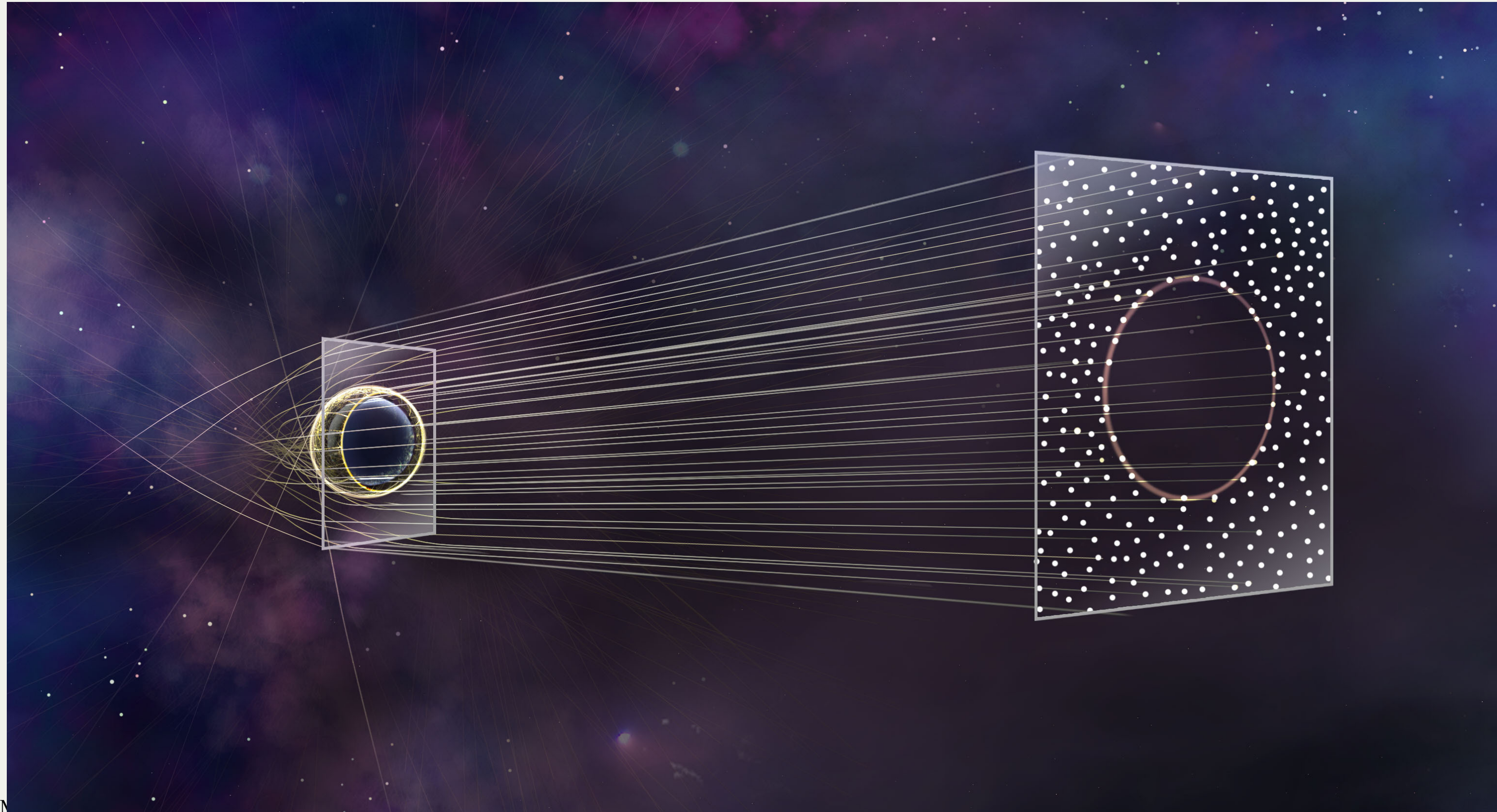


Libraries

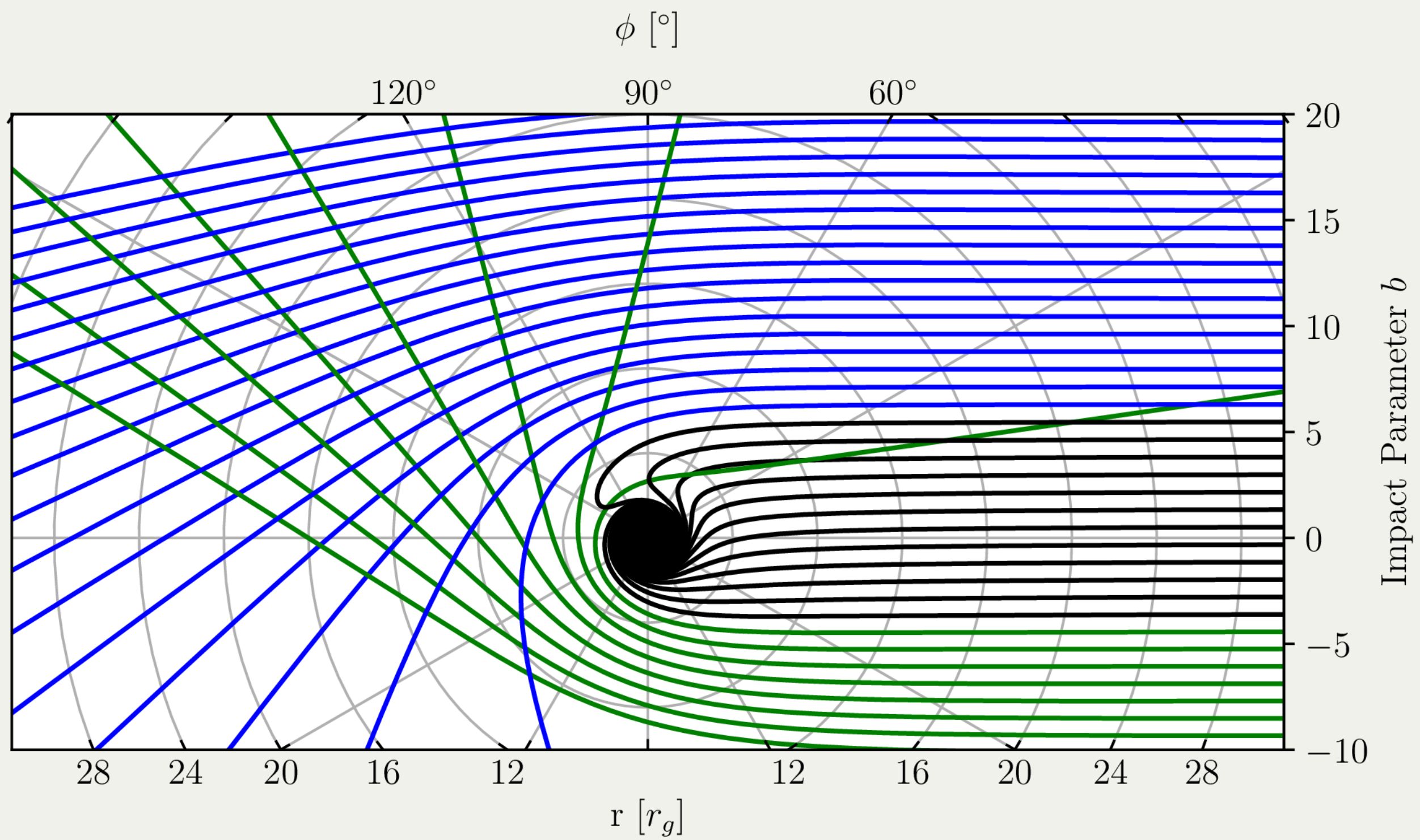
Libraries



The Imaging Process



The Imaging Process



Building Image Libraries

Cutting Image Libraries

Comparisons

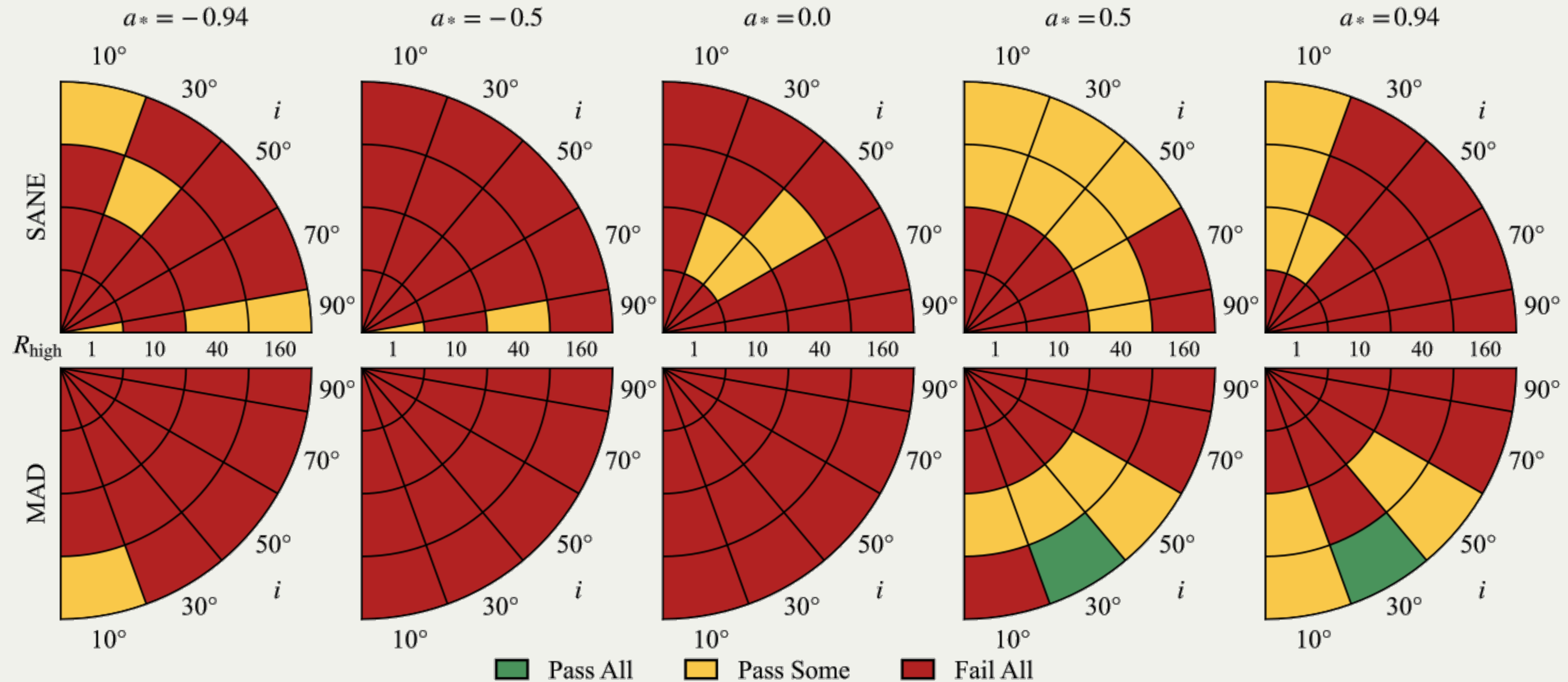
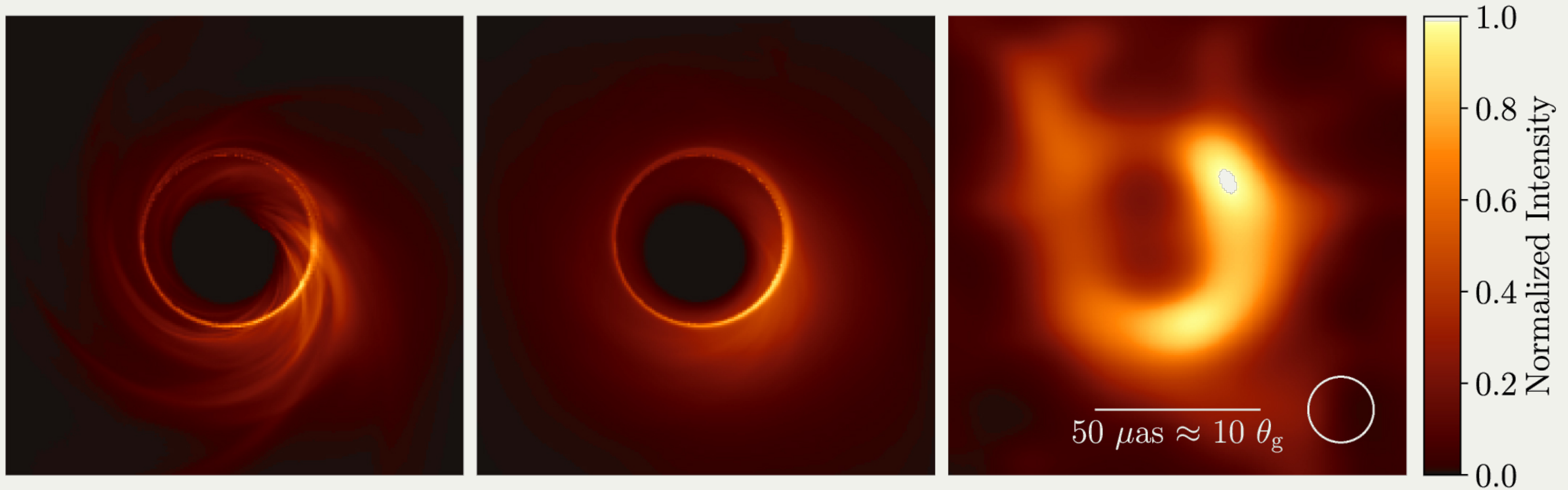


Figure 15. Combined constraints without structural or flux variability. Green indicates that the KHARMA, BHAC, and (for $i = 10^\circ, 50^\circ, 90^\circ$) H-AMR models pass, yellow that one or two of the fiducial models fail, and red that all models fail.

Comparisons



Thanks!

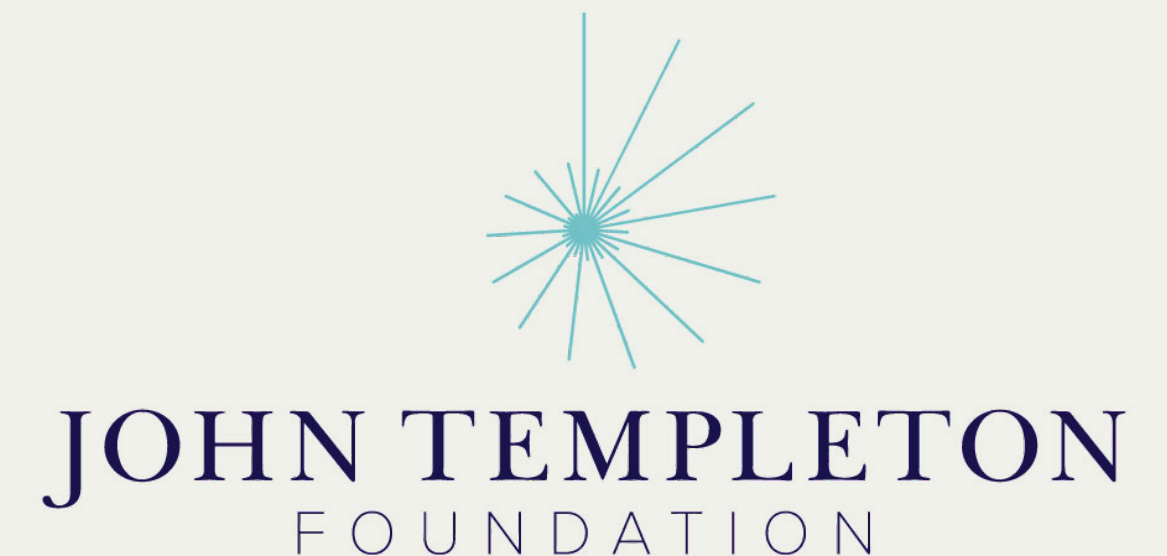
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GORDON AND BETTY
MOORE
FOUNDATION



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